MEGA POWER TECHNOLOGY

Empowering Global Sustainable Energy Development

Integrated Digital Green Energy Solutions and Core Products

MEGA POWER CO., LTD

Japan – Tokyo 2026

Comprehensive Digital Green Energy Solution

Facing global energy shortages and climate change, developing new energy is key to tackling these challenges.

Integrated clean energy utilization enhances resilience, optimizes the energy mix, lowers costs, and supports transportation electrification.

MEGA combines green energy (photovoltaic), stable supply, flexible storage, and efficient charging, powered by digital management tools to deliver a comprehensive green digital energy solution.

Microgrid Energy Storage System Solution

Frequency Regulation Storage Solution

Integrated PV-ESS-Charging Solution

Transportable Energy Storage System Solution

Commercial and Industrial Energy Storage System Solution

Utility-Scale Energy Storage Solution

The Utility-Scale Energy Storage Solution refers to megawatt-level systems deployed on the grid or generation side, aiming to shift energy over time to address the mismatch between power generation and consumption. It is mainly applied in wind farms, solar power plants, and substations or transmission lines nearing capacity limits, helping improve grid stability and flexibility, enhance power reliability, support higher renewable energy integration, and boost overall system efficiency.

Scenarios

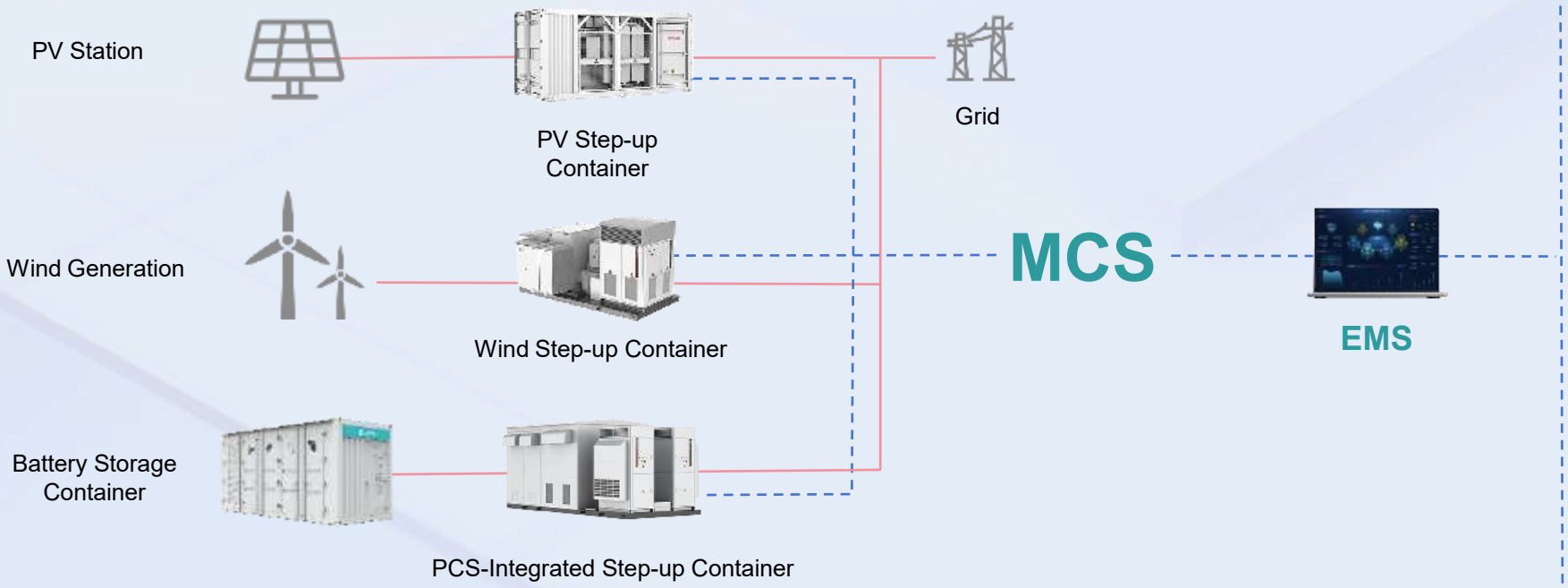
Wind Power Station

PV Station

Transformer Substation

Utility-Scale Energy Storage / Features

- Enhances grid stability and flexibility: With millisecond class active/reactive power regulation
- Enables higher renewable integration: smoothing intermittency and shifting energy
- Provides peak shaving and backup capacity to ease grid congestion and support critical loads
- Improves system resilience with black start and rapid response during faults or extreme events



HCS
Reducing frequency regulation latency more than 50ms

BWS
Gradual failure detected 28 days in advance

EDR
Comprehensive failure data tracing

OMS
Automatic generation of work orders
Realtime progress

Frequency Regulation Storage Solution

Frequency Regulation Energy Storage System Utilizes energy storage technology to regulate power system frequency, ensuring stable grid operation.

Primarily used to address deviations from the target frequency in power systems, it rapidly responds and adjusts active power output to minimize frequency deviations, thereby maintaining grid frequency within the target range. Frequency regulation energy storage stations quickly respond to grid frequency changes, adjusting power output to ensure grid frequency stability around 50Hz.

Scenarios

Grid Frequency Regulation

Green Energy Station

Micro Grid Station

Frequency Regulation Storage / Features

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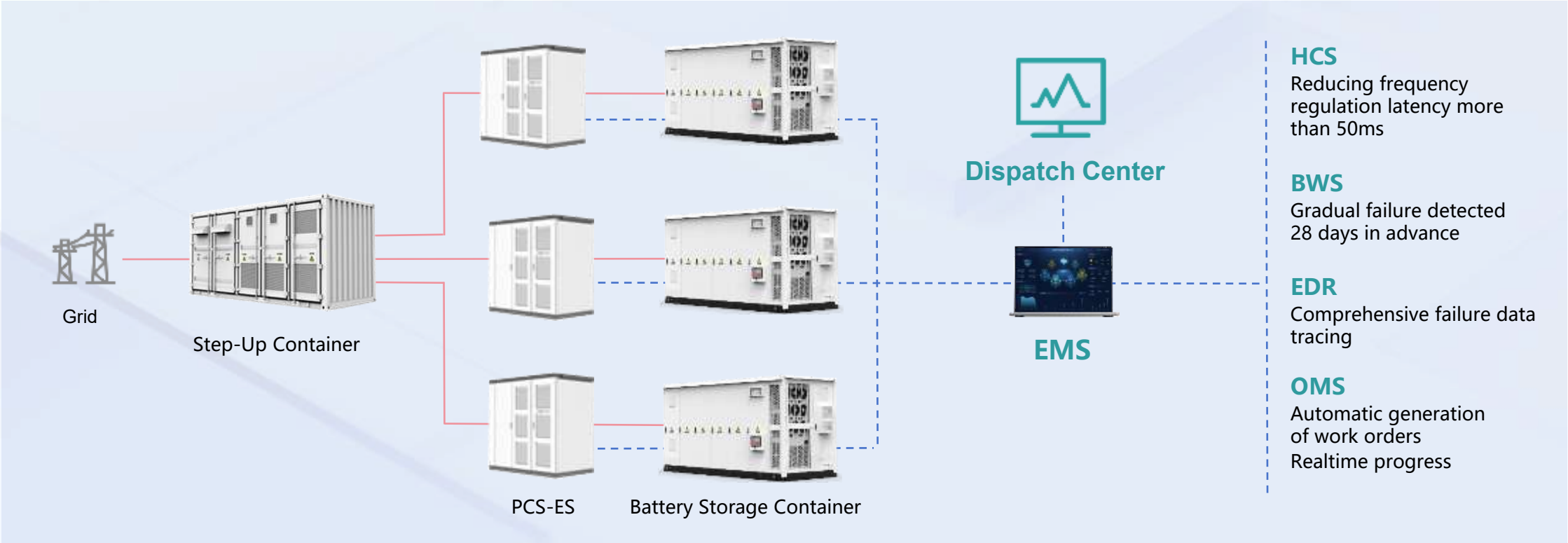
- Instant Respond: Millisecond-level respond to frequency changes or dispatch instructions,surpassing thermal power (Minute-level) and hydro power (Second-level)
- Precise Regulation： Accurately follows instructions for power regulation
- Two-way Regulation capability: Flexibly absorbs or releases power with the grid,.
- Instant Power Delivery: Zero-latency delivery of full power
- Economics: Generate significant revenue in markets



Integrated PV-ESS-Charging Solution

The integrated solution of solar PV, energy storage, and EV charging organically combines three systems—photovoltaic generation, energy storage, and electric vehicle charging—through either AC or DC coupling, forming a smart, efficient, and clean energy microgrid.

This system is well-suited for scenarios with no grid access, weak grids, or limited distribution capacity, meeting the fast-charging needs of passenger cars, buses, heavy trucks, and mining vehicles. According to station scale, energy-storage-integrated chargers can be configured, or the system can be flexibly deployed in modular combinations such as "PV + Storage + Power Cabinet + Battery Cabinet + Ultra-Fast Charging Terminal."

Scenarios

No Grid

Weak Grid

Heavy Truck Supercharge

EV High-Power Charging

Integrated PV-ESS-Charging / Features

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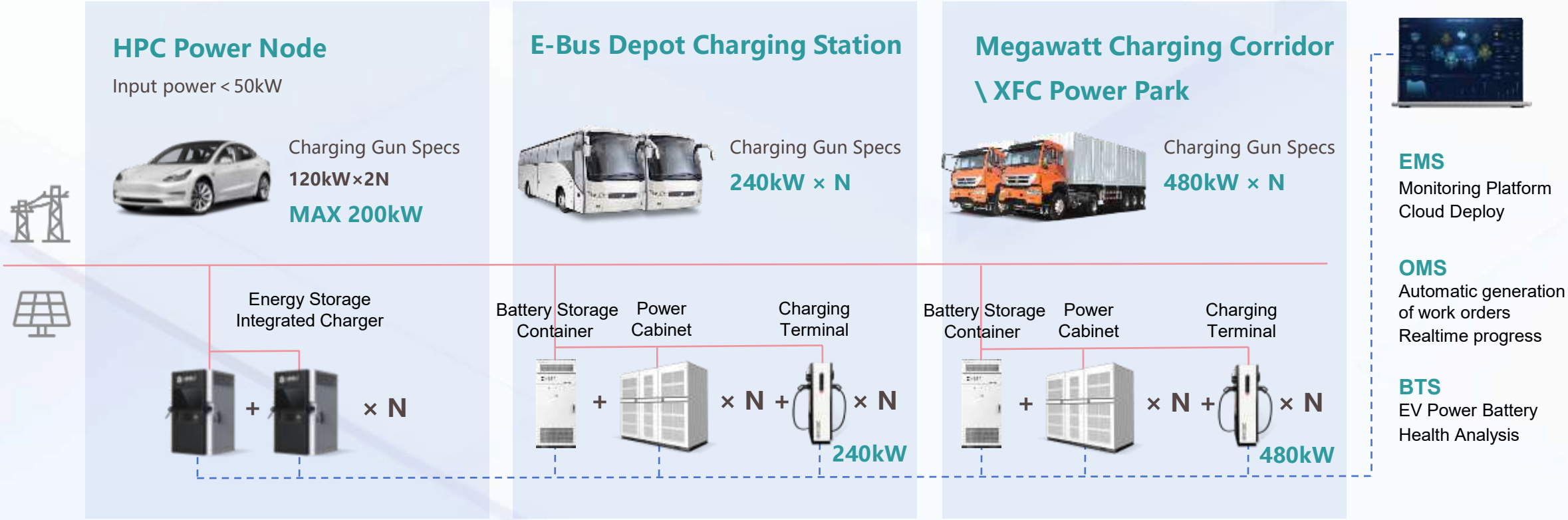
- Virtual Expansion: Auto charge/discharge bypasses grid limits, cuts upgrade costs, and saves fossil fuel
- Stable Supply: Store low-load energy, release at peak; switch Off-grid/On-grid; backup power when Off-grid
- Flexible Deployment: Modular Design with plug-and-use, reintegrated delivery, rapidly deployment
- Smart O&M : Industry-unique "6S+EDR" system with AI alerts, fault recording and fire safety improves maintenance efficiency



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Transportable Energy Storage System Solution

A transportable energy system integrates power generation, storage, and control into a movable platform such as containers, trailers, special vehicles, or vessels—offering rapid deployment and flexible energy supply.

Paired with an intelligent dispatch system, it enables efficient power management and quick response to energy demands, breaking geographical limits to deliver energy "wherever it's needed." Typical applications include emergency response, off-grid and remote power supply, temporary events and construction sites, and grid support in weak areas.

Scenarios

Remote Area

Emergency Rescue

Off-grid Supply

Green Mining

Transportable Energy Storage System / Features

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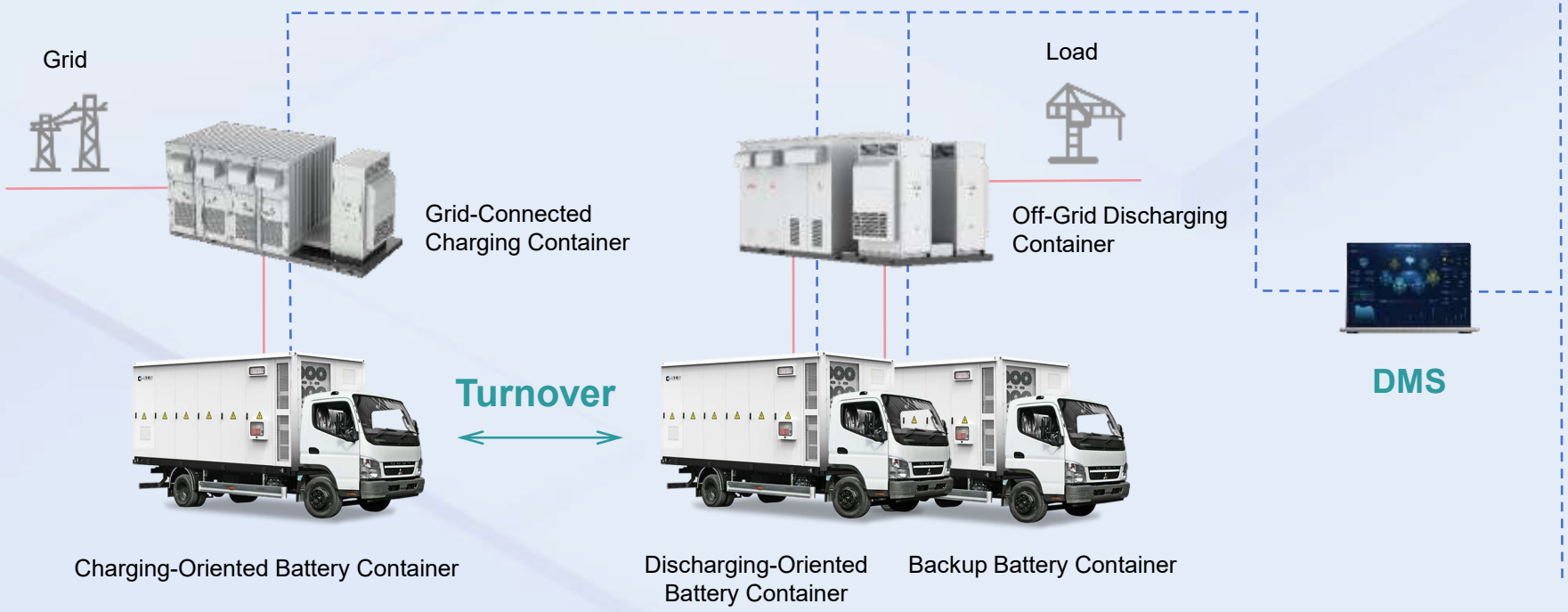
- Fast Plug：Supports 10,000+ mating cycles, ensuring high reliability and flexible protocol configuration.
- Power Supply：Core coponent redundency, seemless battery switching
- Anti-Vibration: Automotive-grade system design with vibration-resistant battery
- Temperature Control：A1 class flameresistant material, all-weather liquid cooling
- Fire Protection System: Off-grid fire suppression power supply with pre-event warning and post-event analysis capabilities
- O&M System: Smart Operations Platform (SOP) enabling automated fault localization and maintenance work order dispatch



SCS
Intelligent stability control algorithmAdapting to various sudden load changes

BWS
Gradual failure detected 28 days in advance

EDR
Comprehensive failure data tracing

OMS
Automatic generation of work orders
Realtime progress

Commercial and Industrial Energy Storage System Solution

Commercial and Industrial energy storage systems are applied in scenarios requiring efficient energy management, cost reduction, or enhanced power stability, such as industrial enterprises, commercial buildings, hospitals, schools, and public facilities, offering flexible configuration, intelligent control, and eco-friendly benefits.

They serve as a core tool for energy cost control, improved power reliability, and green transformation, particularly in high-cost electricity areas, high load fluctuations, or scenarios demanding stability.

Scenarios

- Industiral Enterprise
- CBD
- New Energy Facilities
- School Hospital

Commercial and Industrial Energy Storage System / Features

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- Customized Sizing Service: Capacity assessment & site surveying with tailored ESS layout planning
- Grid Interconnection: Multiple optimization solutions supporting project registration and commissioning compliance
- Safety: Industry-exclusive '6S+EDR' solution ensuring multi-dimensional protection across cell, system, and operational layers
- Smart O&M: Multi-scenario solutions, maximized utilization, data visualization, full data monitoring, and timely fault alerts

Grid

≤ 2MWh

Small-scale Users

0.4kV Input

261kWh Cabinet × N

2MWh~10MWh

Medium-scale Users

10kV Input

Step-up Power Transformer

418kWh Cabinet × N

≥5MWh

Large-scale Users

≥10kV Input

PCS-Integrated Step-up Container

5MWh Containerized ESS × N

EMS
Monitoring Platform
Cloud Deploy

ECS
Intelligent Capacity Caculation for precise decision

EDR
Comprehensive failure data tracing

OMS
Automatic generation of work orders
Realtime progress

“6S+EDR” Innovative System Solution / Product Safety Guarantee

Leveraging over two decades of expertise in power generation and user-side projects, we have pioneered the world's first energy storage power station EDR system, active safety system, and high-speed control system. This solution deeply integrates cutting-edge technologies such as the Internet of the Power Industry, artificial intelligence, and edge computing with advanced energy storage technologies and traditional power system control and protection techniques, significantly enhancing the safety, intelligence, and grid compatibility of energy storage system products.

Battery Warning System
BWS
Data-driven algorithm,
Intelligent advance warning
preventing battery cluster fire

Event Data Recorder
EDR
Multi source data collection
Credible accident tracing
Comprehensive record
Analysis of fault causes

High-speed Control System
HCS
Real-time Ethernet technology,
high speed synchronization and
low delay regulation

Operations & Maintenance System
OMS
Real-time monitoring, work order
generation, and visible progress

Battery Management System
BMS
Connector temperature detection,
PACK-level fire prevention linkage

Power Conversion System
PCS
The series multi-branch design
is effective in electrical and
safety isolation

Energy Management System
EMS
Dynamic Intelligent Control
System-level partition safety
management

EMS On-Premises Deployment and Data Security

/ Data Safety Guarantee

Base on scale, EMS localize deploy targets Small-scale（Under 1MWh）, Medium-scale（1MWh~5MWh）、Large-scale（5MWh above）Three different scales offer various customized EMS deployments and data security solutions. The data security management is designed from several aspects including **data collection security, data transmission security, data storage security, system function security, server system security**, and compliance guarantee.

E2EE

End-to-End Encryption

During the data transmission process, the TLS 1.3 protocol is used, and when storing, the AES-256 encryption algorithm is adopted to ensure that the data will not be illegally accessed at any stage.

RBAC

Role-based Policies Access

The role-based access control mechanism ensures that personnel in different positions can only access the data sets and API interfaces within their respective job responsibilities.

MFA

Multi Factor Authentication

Users need to log in to the system through a combination of password and dynamic verification code/bio-recognition for enhanced account security.

Log Audit

Behavior tracking

All operation records are written into an unalterable log system, and support multi-dimensional audit queries by time, user, operation type, etc.

Digital Energy Management Platform / O&M Safety Guarantee

This platform is comprehensive in its functions and highly efficient in management, and is suitable for the application management of various large, medium and small-scale energy storage systems.

In addition to the industry-exclusive "Event Data System EDR" which has obtained a national invention patent, it also integrates the "Maya Intelligent Power Prediction System ECS" "Energy Management System EMS" "Operation and Maintenance System OMS" and "Battery Warning System BWS"

Comprehensive Data Storage

Visualized Analysis

Intelligent Strategy Dispatch

Precise Early Warning

Safety Analysis

Efficiency in O&M and Monitoring

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PCS-Integrated Step-up Container (Cluster)

/ Micro-grid & Frequency Regulation Storage Core Product

Model	iBCS100-2500B/20-AC
DC Side Parameters	
DC Input Channels	12
DC Maximum Current	236A*12
DC Voltage Operation Range	1000V～1500V
AC Side Parameters	
Power*Channels	215kW*12
Rated Power	2500kW
Maximum Output Power	2750kW
Rated Voltage	20kV
Isolation	Converter Isolate
Grid-Connected Operation Parameters	
Rated Grid Voltage	20kV
Rated Grid Frequency	50Hz/60Hz
Power Factor	- 1～1
System Parameters	
Sizes	6500mm×2900mm×2900mm
Weight	≤18T
Ambient Temperature Range	- 40℃～ + 60℃ （>50℃ Derated）
Relative Humidity Range	0～100%
Maximum Altitude	≤4000m （Not Derated Under 3000m）
Protection Class	IP54 （PCS IP66）
Grid Supply	High/low voltage ride-through function, frequency regulation function, voltage regulation function, grid-forming function

PCS-Integrated Step-up Container (Integrated)

/ Micro-grid & Frequency Regulation Storage Core Product

Model	iBCS100-2500A/20-AC
DC Side Parameters	
DC Input Channels	2
DC Maximum Current	1408A×2
DC Voltage Operation Range	1000V～1500V
AC Side Parameters	
Power*Channels	2500kW
Rated Power	2750kW
Maximum Output Power	2750kW
Rated Voltage	20kV/35kV
Isolation	Converter Isolate
Grid-Connected Operation Parameters	
Rated Grid Voltage	20kV/35kV
Rated Grid Frequency	50Hz/60Hz
Power Factor	- 1～1
System Parameters	
Sizes	6058mm×2438mm×2896mm
Weight	≤18T
Ambient Temperature Range	- 40℃～ + 60℃ （>50℃ Derated）
Relative Humidity Range	0～100%
Maximum Altitude	≤4000m （Not Derated Under 3000m）
Protection Class	IP54 （PCS IP66）
Grid Supply	High/low voltage ride-through function, frequency regulation function, voltage regulation function, grid-forming function

DC Battery Storage Container

/ Micro-grid & Frequency Regulation Storage Core Product

Model	iESS100-5015B2-LC
Battery Parameters	
Battery Type	LFP
Patterns	416S12P
Battery Capacity	5015.9kWh
Battery Rated Voltage	1331.2VDC
Battery Voltage Range	1164.8~1500V
System Parameters	
Battery System Sizes	6058mm×2438mm×2896mm
Battery System Weight	42000 kg
Protection Class	IP54
Ambient Temperature Range	-20 ℃ ~ 45 ℃
Battery Thermal Management	Liquid-cooling/heating
Fire Extinguishing System	Combustible Gas Detection + Emergency Ventilation + Gas Fire Suppression System + Water-based Fire Protection System
Communication Protocol	CAN, RS485, TCP/IP

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PV-Storage Hybrid Unit

/ Integrated PV-ESS-Charging Core Product

Model	iPES100-100-215-AC-100
AC Side Parameter(Grid-Connected)	
RatedPower	100kW
Rated Grid Power	AC 230/400V
AC Input Method	3L/N/PE
Grid Frequenct Range	50±2.5Hz
Grid Voltage Range	-10% ~ +10%
Battery Parameter	
Battery Type	LFP
Capacity	215.04kWh
Rate Voltage	DC 672V～876V
PV Input Parameter	
Maximum Input Power	2×50kW
PV Maximum Open-Circuit Voltage	DC 800V
MPPT Operation Voltage Range	DC 400～650V
System Parameter	
Efficiency	Under Rated Status≥86%
Operation Ambient Temperature	-20℃ ~ +45℃
Protection Classes	IP55
Weight	3.0T
Sizes	1800mm×1250mm×2300mm (Width×Depth×Height)
Authentication	GB/CE

Energy Storage Integrated Charger

/ Integrated PV-ESS-Charging Core Product

Model	iEFC100-80/120-AL-C
AC Side Parameter	
Rated Power	120kW
Rated AC Input	AC 400V
AC Input Method	3L/N/PE
Grid Frequency Range	50±2.5Hz
Grid Voltage Range	-10% ~ +10%
DC Side Parameter	
Capacity	186kWh / 209kWh
Rated DC Power	120kW
System Parameter	
Charging Gun A Max Power	120kW(Max200kw)
Charging Gun B Max Power	80kW(Max200kw)
Output Voltage Range	DC200~1000V
Max Output Current	250A
Operation Temperature	-25℃ ~ +45℃
Noise Level	≤75dB
Protection Classes	IP55
Sizes	1350mm×1300mm×2300mm (Width×Depth×Height)

Integrated PV-ESS-Charging Power Cabinet & High-Power Charging Terminal

/ Integrated PV-ESS-Charging Core Product

Model	iEFC
Power Cabinet Parameter	
Rated Power	1440kW
Rated Input Power	200Vdc～850Vdc
Rated Output Power	50Vdc～1000Vdc
Maximum Output Channel	12
Charging Terminal Parameter	
Rated Output Voltage	200Vdc～1000Vdc
Rated Output Ampere	2×600A+10*250A
System Parameter	
Operation Temperature	-25℃ ~ +45℃
Power Cabinet Weight	2.0T
Charging Terminal Weight	270kg
Protection Classes	IP55
Power Cabinet Sizes	2300mm×1100mm×2100mm (Width×Depth×Height)
Charging Terminal Sizes	500mm×600mm×1800mm (Width×Depth×Height)

Grid-Connected Charging Container

/ Transportable Energy Storage System Core Product

Model	iBCS100-1725A/10-AC
DC Side Parameter	
DC Side Max Current	1807A
DC Voltage Operation Range	1000V ~ 1500V
AC Side Parameter	
Power*Channel	1725kW
Rated Power	1725kW
Max Output Power	1898kW
Rated Voltage	10kV
Isolation	Converter Isolate
Grid-Connected Operation Parameter	
Rated Grid Voltage	10kV
Rated Grid Frequency	50Hz/60Hz
Power Factor	- 1 ~ 1
System Parameter	
Sizes	6058mm×2438mm×2896mm
Weight	≤13T
Ambient Temperature Range	- 40℃ ~ + 60℃ （>50℃ Derated）
Relative Humidity Range	0 ~ 100%
Maximum Altitude	≤4000m （Not Derated Under 3000m）
Protection Classes	IP54 （PCS IP66）

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Off-Grid Discharging Container

/ Transportable Energy Storage System Core Product

Model	iBCS100-2×2500A/10-AC
DC Side Parameter	
DC Side Max Current	2816A
DC Voltage Operation Range	1000V～1500V
AC SIde Parameter	
Power*Channel	2500kW × 2
Rated Power	2510kW
Max Output Power	2761W
Rated Current	2100A
Grid-Connected Operation Parameter	
Rated Grid Voltage	10kV
Isolation method	Converter Isolation
Rated Grid Frequency	50Hz/60Hz
Power Factor	- 1～1
Off-grid Operation Parameter	
Rated Output Voltage	0.69kV
Rated Output Power	2500kW
Rated Output Frequency	50Hz/60Hz
System Parameter	
Sizes	7200mm×3000mm×2896mm
Weight	≤18T
Protection Classes	IP54 （PCS IP66）

Battery Storage Transport Vehicle

/ Transportable Energy Storage System Core Product

Model	iESS100-3343B2-LC-M
DC Side Parameter	
Battery Type	LFP
DC Side Rated Voltage	1331.2V
Operation Voltage Range	1164.8V ~ 1500V
Rated Charge/Discharge Power	1725kW
System Battery Cycles	6000（95DOD%Charge/Discharge，25℃，EOL80%）
System Parameter	
Efficiency	88%
Thermal Management	Variable-frequency Liquid-cooling
Liquid-cooling Capacity	40kW
Fire Extinguishing System	FM200 Fire Suppression + Water Sprinkler Fire Protection
Unique Design	Anti Vibration
Sizes	6058mm×2438mm×2896mm
Weight	≤40T
Maximum Altitude	≤2000m
Protection Classes	IP54
Anti Corruption Classes	C4

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Distributed Energy Storage Cabinet

/ Distributed Energy Storage Core Product

Model	iDPS100-125/261 LC	iDPS200-200/418 LC
AC Parameter		
Rated Grid Voltage	AC380V	AC690V
Rated Output Volage	50Hz/60Hz	
AC Input Method	Three-phase Four-wire	
Rated Charge/Discharge Power	125kW	200kW
DC Side Parameter		
Battery Type	Lithium-ion Battery (LFP)	
Rated Capacity	261kWh	418 kWh
DC Voltage Range	728V ~ 949V	1164.8V ~ 1500V
Rated Charge/Discharge Power	125kW	200kW
Efficiency/Features/Safety		
Efficiency	≥88%	
Features	Anti-backflow Protection、Overload Protection、Scheduled Curve、Anti-islanding Protection、Load Tracking、Demand Control、Remote Dispatch and Monitoring， OTA Remote Upgrade。	
Safety	Overvoltage, undervoltage, short circuit, overload, overcurrent, overtemperature, lightning protection, leakage protection, and full-cabinet plus PACK-level immersive fire protection design.	
System Parameter		
Sizes(Width/Depth/Height)	1000*1350*2480mm	1300*1350*2480mm
Weight	≤3200kg	
Altitude	＜2000m（100%AC Output）	
Thermal Management	Liquid-cooling	
Protection Classes	IP55	
Anti Corruption	C4	

O&M Strategy

Localized Service Team

Establish an operation and maintenance service team in Europe to better understand the local market and customer needs, and provide customized services

Cross-Cultural Communication Training

Conduct cross-cultural communication training for the service team to ensure that team members can effectively communicate with customers from different cultural backgrounds.

Continuous Technical Support and Updates

Provide continuous technical support services and regularly update technical solutions to meet the constantly changing market demands.

Expansion of the Partner Network

Establish partnerships with local enterprises, leveraging their market knowledge and resources to jointly provide more comprehensive services.

Service Mechanism

Emergency Fault Resolution

Set up 7×24h Emergency Hot line, Ensure rapid response to after-sales service requests

Maintenacne

Establish a periodic inspection and maintenance plan to prevent system failures and ensure service continuity through preventive maintenance.

Customer Training and Support

Offer online and on-site training to ensure that customers can use the product correctly and reduce the service demands caused by incorrect operations.

Replacement Part Rapid Delivery

Establish a replacement parts warehouse network to ensure that critical components can be quickly delivered within 24 hours in case of damage, thereby reducing the repair time.

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THANKS !

Continuous innovation in technology drives high-efficiency and secure utilization of clean energy.